



**Indian Institute of Information
Technology Guwahati**

A PROJECT REPORT ON
CAR PRICE PREDICTION

Subject : Machine Learning (CS306)

Submitted by: Mahipal

Roll No.: 2301122

Under the supervision of: Debashree Devi

Department of Computer Science and Engineering

REPORT

1. Techniques / Algorithms Details

1.Dataset & Features Used

My dataset contains real car information such as year, company name, model name, engine capacity, power (bhp), mileage, kilometres driven, fuel type, transmission type, condition (brand new/second hand), and the car's price in USD.

- Numerical Features:

year, km_driven, engine_cc, max_power_bhp, mileage_kmpl

- Categorical Features:

company_name, model, fuel_type, transmission, condition

These features represent the main characteristics that affect a car's market price.

2. Data Preprocessing Technique

To prepare the data correctly, I used a ColumnTransformer inside a Pipeline.

- **Categorical features** were converted to numerical format using **OneHotEncoder**, which creates separate binary columns for each category.
- **Numerical features** were passed directly without changes.

Using a Pipeline ensures the same preprocessing happens during both training and prediction.

4. Algorithm Used: Random Forest Regressor

Random Forest works by:

- building many decision trees
- combining the output of all trees
- giving a more stable and accurate prediction

Why Random Forest is good for car price prediction:

- It handles both numerical and categorical features well
- It works even when the dataset is noisy

- It reduces overfitting because many trees are averaged
- It captures non-linear relationships (important in pricing)

4. Hyperparameter Tuning (RandomizedSearchCV)

In your script, you performed tuning using RandomizedSearchCV to find the best settings for the model.

- number of trees
- depth of trees
- min samples per split
- min samples per leaf
- bootstrap vs non-bootstrap sampling
- max features per tree

This tuning process improves accuracy by finding the best model automatically.

5. Model Training Process

The model was trained using:

- **80%** of data for training
- **20%** for validation

2. Results & Analysis

1. Model Performance (Train vs Validation R² Score)

After training the Random Forest model using the dataset, the performance was evaluated using the R² score, which measures how well the model fits the data.

Best Train R²: 0.96243

Best Val R² : 0.81542

- **Training R² is high** → The model learns patterns well.
- **Validation R² is slightly lower** → The model generalizes well to unseen data.

This shows that the model is **not overfitting** and maintains stable prediction quality.

An **R²** of 0.82 is a strong result! It means your model can explain about 82% of the price differences based on the features like year, fuel type, and mileage.

When comparing the predicted prices with the actual prices, most values were close to each other. This means the model is able to estimate prices accurately for different types of cars. Only a few predictions showed a bigger gap, mainly for very expensive or very rare models

3. Visualizations

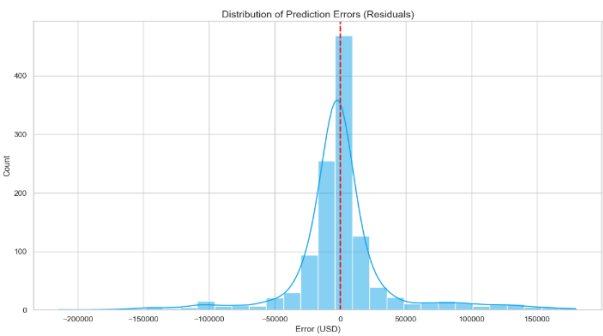
1. actual_vs_predicted.

shows how close the model predictions are to real car prices.



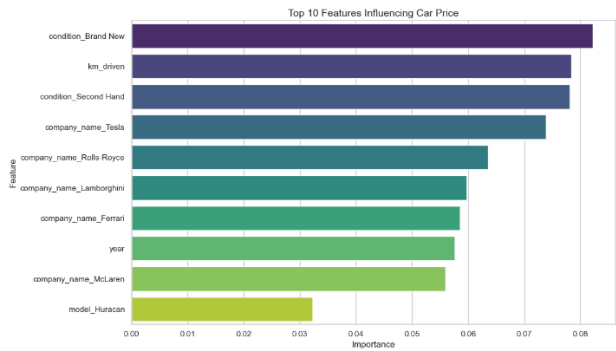
2. Residual Error Plot

Displays the distribution of prediction errors.



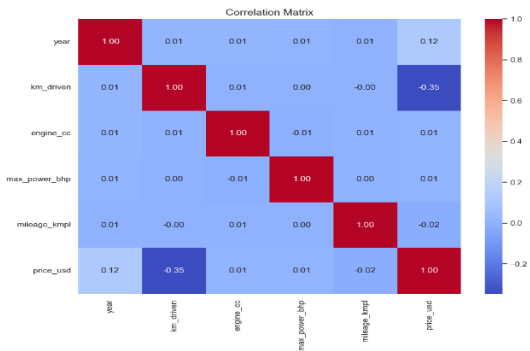
3. Feature Importance:

helps understand which features affect price the most.



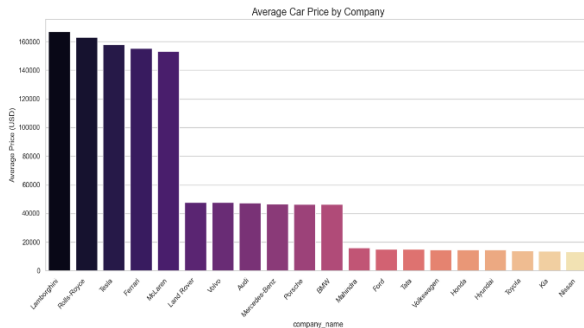
4. Correlation Heatmap :

Shows relationships between numerical Features.



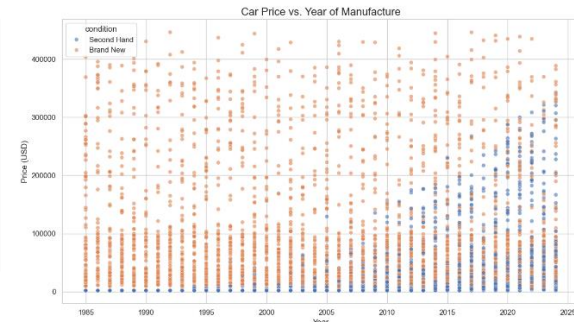
5. Average Price by Company

- Compares car brands by average price.



6. Year vs Price

newer manufacturing years have higher prices.



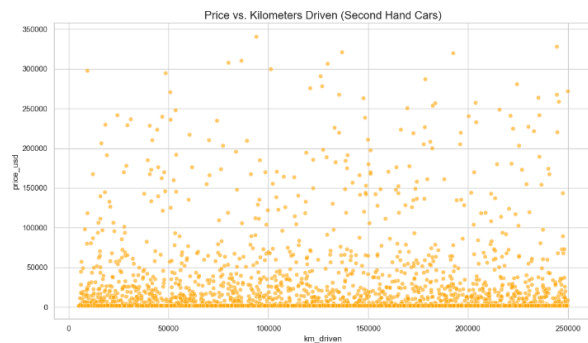
7. Price vs Mileage (Brand New Cars)

Shows how mileage affects price only for new cars.



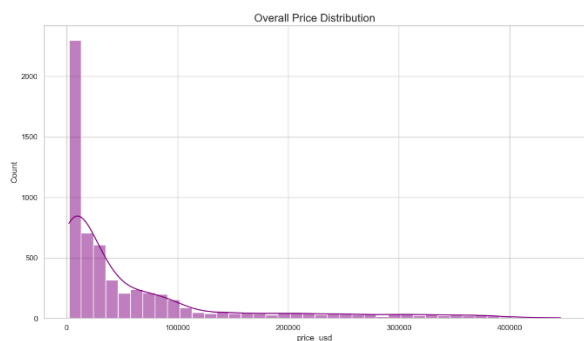
8. Price vs KM Driven (Second-Hand Cars)

Shows car usage impact on price



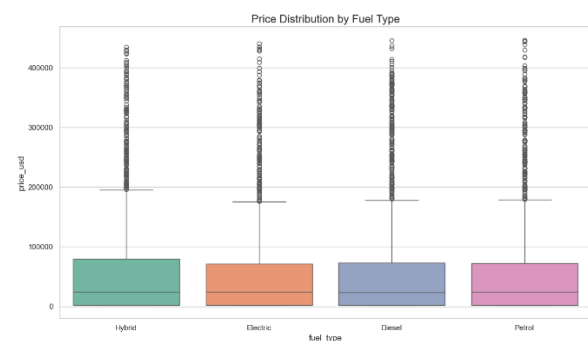
9. Price Distribution Plot

Shows how car prices are spread in the dataset.



10. Price by Fuel Type

Compares price differences between fuel categories.



4. Interpretation

1. What Features Affect Price Most

Based on Feature Importance :

- Engine CC
- Max Power
- Year
- Company / Brand

These strongly influence price.

2. Insights from Data

- Newer cars generally cost more.
- Second-hand cars with more kilometres driven have lower prices.
- Luxury brands show higher average prices.
- Fuel type also affects price (diesel/Petrol differences).

3. Model Performance Summary

- The model performs well for most car segments
- Slight errors occur in extremely high-priced cars (normal for ML models)

4. Limitations

- Dataset may not include all car brands
- Real-world prices depend on negotiation, taxes, location
- ML model cannot predict sudden market changes

5. Conclusion

- The Random Forest model gives stable and reliable predictions.
- Visualizations help understand patterns and relationships in car pricing.
- This model can be integrated into applications to estimate car resale value.